

ARCHITECTURAL FREEDOM (SPECIFICATION STRIPPED AWAY)

MAXIMAL

MINIMAL

(c) AI-Native Design System (Layer 3)

NO IMMUTABLE STRUCTURAL TEMPLATE

Input Specification
(Specification Stripped Away)

★ System Intent & Golden Spec

- Interface Protocols
- Pipeline Topology
- Memory Hierarchy
- Arithmetic & Datapath

Human sets intent and contract • System proposes design

ZERO TEMPLATE: Cross-Layer Co-Design Synthesis Fabric

Compiler / Schedule

Custom Vector ISA

Carry-Save ALU (Redundant)

Streaming FIFO Queues

Unconstrained pint HW/SW, arithmetic, and interface co-adaptation

Physical Signoff Oracles
(Silicon Ground Truth)

STA • DRC/LVS • LEC/FPV
CDC • Dynamic IR-Drop
Multi-fidelity physical verification

Signoff Before Commitment
Candidates must pass signoff before human commitment

Container dissolved: cross-layer co-adaptation across SW and silicon ↑

(b) Container-Bound Optimization (Layer 2)

PARAMETER SEARCH IN IMMUTABLE CONTAINER (Θ)

Input Specification
(Locked Template Bounds Θ)

- Workload & PPA Bounds
- Interface Protocols
- Core Pipeline Topology
- Subsystem Dimensions
- Component Floorplan

Topologies frozen • Parameters Θ automated

LOCKED TEMPLATE (Θ) • Swept Knobs (Θ)

512KiB 4MiB
Cache Cap. Θ
[1 MiB]

4 64
Buffer Depth Θ
[32 entries]

SRAM
Macro Floorplan
 $\Theta = (x^*, y^*)$

Trapped by Memory Bandwidth Wall & Routing Congestion

Closed-Loop Optimizer
(Fast Heuristic Surrogates)

Cycle Sim & Fast EDA Surrogates
RL / Bayesian sweep (AlphaChip)
Evaluates scalar surrogate PPA proxy

Local Optimum Θ^*
Bounded by template perimeter

Parameters automated: numerical optimization within frozen topology ↑

(a) Point Assistance (Layer 1)

LOCALIZED POINT ASSISTANCE IN FROZEN LEGACY CONTEXT

Input Specification
(Fully Human-Frozen)

- Workload & PPA Bounds
- Interface & Bus Protocols
- Core Pipeline Topology
- Subsystem Dimensions

★ AI Prompt: "Draft Leaf RTL"
Human specifies the architecture • Localized leaf query

LOCKED: Human-Frozen Architecture (Rigid SoC)

Fetch
AI Leaf

Decode

Execute

Mem

WB

Fixed Interfaces • Frozen Microarchitecture • Open-Loop

Human Verifier
(Manual Integration Tax)

Syntactic Candidate
Drafted Verilog without proof
Potential CDC & timing flaws

Manual Integration Tax
Human rewiring, lint & signoff